EECS 461: Embedded Control Systems, Winter 2025

CLASS TIME: Tuesday and Thursday, 9:00AM–10:30AM.
PLACE: 1500 EECS (lecture) 4342 EECS (lab)

INSTRUCTOR: J.A. Cook
OFFICE: 2215 EECS
EMAIL: jeffcook@umich.edu
OFFICE HOURS: Tuesday and Thursday from 10:30 AM–12:30 PM (EECS 2215). Don't hesitate to schedule an appointment, or email me any time.

LAB INSTRUCTORS:

Michael Sun michaesu@umich.edu, (Sections 11 and 18, Monday 3:00-6:00 and Friday 10:00 AM-1:00)

Gabriel Lounsbury lounsbg@umich.edu, (Sections 16 and 17, Thursday 11:00-2:00 and 3:00-6:00)

Isabelle Nolan inolan@umich.edu, (Sections 14 and 15, Wednesday 10:00-1:00 and 3:00-6:00)

Zachary Coneybeare zjcbeare, (Section 12, Tuesday 11:00-2:00)

Jake Pattok jpattok@umich.edu, (Section 13, Tuesday 3:00-6:00)

- Labs will begin **Monday, January 13**.
- Everyone will be assigned a lab partner.
- Be sure to read and understand the lab collaboration policy and late-lab assignment policy at the end of this syllabus and posted on CANVAS.

GRADING:

- Homework: 30%
- Laboratory Assignments: 30%
- Exam(s): 20%. There will be one take-home exam tentatively the week beginning March 10.
- Project: 20%.

HOMEWORK is to be submitted **electronically** as specified in the CANVAS Assignment. You will have at least a week to complete each assignment, thus **late homework will not be accepted.** The *Homework Policy* and *Lab Policy* are posted on the CANVAS website, and included in the syllabus. Please read them and follow instructions.

ATTENDANCE: Students are expected to ALL labs from BEGINNING to END. Lectures will be recorded and posted to Canvas. **Be sure to inform your lab instructor and your lab partner if you must miss lab for any reason (illness, for example) so that you can arrange open lab time to complete your assignment.**

TEXTBOOK: There is no required textbook for this course. Lecture notes and other useful information will be posted on Canvas.

OTHER USEFUL REFERENCES:

- David E. Simon, *An Embedded Software Primer*, Addison Wesley, 1999, ISBN: 0-201-61569-X.
- J. Lemieux, *Programming in the OSEK/VDX Environment*, CMP Books, 2001.
- D. Auslander and C.J. Kempf, *Mechatronics: Mechanical Systems Interfacing*, Prentice-Hall, 1996.
- J. Ledin, *Embedded Control Systems in C/C++*, CMP Books, 2004.
- A. Burns and A.J. Wellings, *Real-time Systems and Programming Languages*, Pearson, 2001.
- C.M. Krishna and K.G. Shin, *Real-time Systems*, McGraw-Hill, 2001.

OVERVIEW: The vast majority of microprocessors are not used in desktop or laptop computing applications. Instead, they are embedded in other technological systems, such as cell phones, appliances, and automobiles. One unique feature of embedded applications is that the goal of the design is not directly related to the performance of the microprocessor, but rather to the performance of the overall system. When I drive my car, I don't care what kind of microprocessors is being used to control the engine, transmission, and brakes; I only care that they work, and work reliably. In EECS 461 you will learn how to use a microprocessor as a component of an embedded control system.

The specific embedded system we will be working with is a haptic interface, or force feedback system. The skills we shall develop are applicable, however, to a broad range of embedded system applications.

Lectures

The lectures cover a wide range of topics, including:

- Sampling. Position and Velocity Measurements. Encoders. Quadrature Decoding.
- The NXP S32K144 FlexTimer and its Quadrature Decoding function.
- Features of the NXP S32K144 Microcontroller.
- Pulse Width Modulation (PWM). Frequency response of PWM signals. DC motors. Amplifiers.
- Interface electronics.
- Haptic interfaces. Virtual worlds. Human-computer interaction.
- Artifacts due to microprocessor implementation of the virtual world.
- Algorithms. Feedback control. Logic control and finite state machines. Numerical integration.
- Implementing a virtual world on a microprocessor.
- Concepts from real time operating systems. Interrupts. Shared data. Latency. Software architecture.
- Modeling. Use of Matlab/Simulink/Stateflow to simulate the interaction of a virtual world with a human operator through the haptic interface.
- Networking. Control Area Network (CAN) vs. Ethernet protocol. The CAN unit on the NXP S32K144.
- Rapid prototyping and automatic code generation.

Laboratory You will have a lab partner. During the first several weeks of the semester, the laboratory exercises will develop an embedded controller for a haptic interface. The software will be written in C. We will implement the controller over a CAN network to study performance degradation due to networking delay. We will then recreate our work using rapid prototyping tools to generate C code directly from a Simulink model of the haptic virtual world. Late in the semester we will complete a driving simulator project using the hardware, software, and haptic interfaces in the embedded systems laboratory.

Note:

- You must attend the entire 3 hour lab section for which you have registered.
- You may not change lab sections.

Lab Policies

- **Late arrival: Students will be penalized 20% of the lab grade for every 10 minutes they are late. Please plan to be on time!**
- Prelabs and postlabs: Prelabs will be due online at midnight the day BEFORE your lab meets. Postlabs will be due online at midnight the day BEFORE your lab meets the following week.

Homework Policy

(1.) Homework is due electronically as specified in the CANVAS Assignment. Late homework will not be accepted.

(2.) You may consult with one another on homework problems.

(3.) The solutions you hand in must be written up by yourself, and represent your own intellectual output.

(4.) **Under no circumstances may you look at solutions from previous years. Nor may you use Matlab and Simulink code from previous years. It is a violation of the Honor Code to do so.**

Lab Collaboration Policy

(1.) Pre-lab questions: These are questions asked about the concepts that the lab addresses, and often relate to topics covered in lecture. Some pre-lab questions also test your knowledge of the NXP S32K144 and its peripherals. Such questions can be answered by referring to the relevant sections of the NXP S32K144 manual or to any programming notes that accompany a lab. All of these questions are to be answered **INDIVIDUALLY**.

(2.) Pre-lab code, for use in lab: Some labs will come accompanied with header files and skeleton library files and instructions about how certain functions should behave. These functions will be used during the in-lab, and you can, and should, work with your partner on writing them before coming to lab.

(3.) In-lab experiments: You will work together with your partner on anything in the "In-Lab" section of a lab handout.

(4.) Post-lab questions: These are questions about any concepts you would have learned about during your In-lab, or inference questions to test your understanding of what you observed in lab. You may discuss any concepts for these questions with your lab partner, however with **NO ONE** outside your group. You and your partner are required to write up your own solutions **SEPARATELY FROM ONE ANOTHER** for all of the post-lab questions.

(5.) Code to be submitted with the post-lab: You are required to submit, as part of your post-lab, a copy of your code. You should turn in your own copy of the code along with your answers to the pre-lab and post-lab questions.

(6.) **You may not, under any circumstances, with respect to the laboratory exercises or the final project, refer to code or models written by students in previous semesters. It is a violation of the Honor Code to do so.**

Lab Late Policy

(1.) Late pre-labs will not be accepted.

(2.) In-labs are meant to be completed in your assigned section. If you do not complete the in-lab assignment in your assigned section, then you will have one week to complete it during open lab hours.

(3.) It is the student's responsibility to notify the GSI if he or she will miss a lab section.

(4.) Late post-labs will not be accepted.

Reading Assignments There are no required textbooks for this course, but there is a lot of reading material that support the lectures and laboratory assignments. Unless otherwise specified, you will find this material on Canvas: EECS 461 001 W 2024 Files/Lecture Notes.

Links to all the required reading material and a schedule of what you should be reading and when is posted on the CANVAS home page. Make sure you don't fall behind in the reading material–it's important!